

Appendices

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.1 Sensitivity of the Utility Signal

At iteration t , the algorithm computes a per-record utility distribution $g^{(t)}(x_j) \in \mathbb{R}^N$ for each private record x_j , and then averages these utilities over the private dataset $\bar{g}^{(t)}$. Since each $g^{(t)}(x_j)$ lies on the probability simplex, its ℓ_2 norm is bounded by 1. Replacing a single private record can therefore change the average utility by at most $1/m$ in ℓ_2 norm.

It's worth noting that we omit gradient clipping in our setting. Since we compute the average of per-sample gradients, and the utility distribution of each individual record lies within a probabilistic simplex, the ℓ_2 norm of the aggregated gradient is naturally bounded within 1. For any pair of neighboring datasets $\mathcal{D} = \{x_1, \dots, x_{m-1}, x_m\}$ and $\mathcal{D}' = \{x_1, \dots, x_{m-1}, x'_m\}$ differing in a single record, the two averages differ only in one term. Consequently, the ℓ_2 norm between the released statistics is bounded by

$$\left\| \bar{g}^{(t)}(\mathcal{D}) - \bar{g}^{(t)}(\mathcal{D}') \right\| \leq \frac{1}{m} \left(\left\| g^{(t)}(x_m) \right\|_2 - \left\| g^{(t)}(x'_m) \right\|_2 \right) \leq \frac{2}{m}$$

.2 Tree-Based Continual Release: Sensitivity and Privacy analysis

We prove Theorem 1 and provide the corresponding derivation process to ensure that our algorithm satisfies zero-set differential privacy.

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PROOF. theorem 1 is ρ -zCDP, with $\sigma = \frac{1}{m} \sqrt{\frac{2 \log T}{\rho}}$.

We formalize the tree-based aggregation mechanism and its privacy guarantees. Let $\{\bar{g}^{(t)}\}_{t=1}^T$ be the sequence of update signals. We construct a complete binary tree over T iterations: leaves correspond to individual rounds, and internal nodes store interval sums. Each update $\bar{g}^{(t)}$ is incorporated into at most $\lceil \log T \rceil$ tree nodes.

Each tree node v stores a noisy sum:

$$\tilde{S}_v = S_v + \mathcal{N}(0, \sigma^2 T)$$

where S_v is the sum of updates assigned to node v . The released update at iteration t is

$$\tilde{Z}^{(t)} = \sum_{v \sim \text{covers}[1,t]} \tilde{S}_v$$

which uses $O(\log T)$ nodes.

Then, the ℓ_2 sensitivity of the released signal is bounded by:

$$\Delta_{tree} \leq \sqrt{\log T \cdot \left(\frac{2}{m} \right)^2} = \frac{2}{m} \sqrt{\log T}.$$

For the Gaussian mechanism under zCDP:

$$\rho = \frac{\Delta_{tree}^2}{2\sigma^2} = \frac{2 \log T}{m^2 \sigma^2}.$$

To achieve ρ -zCDP, the noise variance is set to

$$\sigma = \frac{1}{m} \sqrt{\frac{2 \log T}{\rho}}.$$

The entire sequence $\{\tilde{Z}^{(t)}\}_{t=1}^T$ satisfies ρ -zCDP. By post-processing, the final reweighted distribution and synthetic dataset preserve the same privacy bound. $\square$

.3 Error analysis

As can be seen from Algorithm ?? (DP-CRW), the generation of the final synthetic dataset $\tilde{\mathcal{D}}$ is determined by the weights $w^{(T)}$ of each sample in the candidate set $\mathcal{C}$, and these weights gradually fit the true distribution of the privacy dataset $\mathcal{D}_{\text{priv}}$ through T rounds of iterations. Specifically, the weight update in each round depends on the noisy prefix sum $\tilde{Z}^{(t)}$ released by the privacy-preserving mechanism (TreeAggregator), which directly determines the direction and magnitude of the weight update. Therefore, in conducting

the overall error analysis, we first measure the bias and error introduced by the noisy prefix sum $\tilde{Z}^{(t)}$ compared with the true utility signal (i.e., the true gradient $\mu = \mathbb{E}[g^{(t)}]$), and then substitute this error term into the weight iterative update process to further derive and calculate the final overall error of the algorithm.

Further, we observe that the noisy prefix sum $\tilde{Z}^{(t)}$ at each iteration of the algorithm is generated from the empirical utility signal $g^{(t)}$ of the corresponding round, with noise subsequently injected through the tree-aggregation privacy mechanism. Based on this generation pipeline and the empirical risk decomposition framework in statistical learning theory, we decompose the total error of $\tilde{Z}^{(t)}$ relative to the true utility signal μ into two independently analyzable components: the empirical estimation error, and the stochastic noise perturbation from the differential privacy mechanism. The specific formulation of each error term is presented as follows:

$$\tilde{Z}^{(t)} = \underbrace{\mu}_{\text{true signal}} + \underbrace{e_{\text{est}}^{(t)}}_{\text{estimating}} + \underbrace{e_{\text{DP}}^{(t)}}_{\text{noise}}$$

Next, we first rigorously derive and upper-bound each of the three aforementioned error components separately, and then substitute them into the exponential refinement iteration process to derive and quantify the overall cumulative error introduced by the weight update procedure.

3.1 Error from estimating.

LEMMA .1. *Let $\delta > 0$. Then with probability at least $1 - \delta$, the sampling error*

$$e_{\text{est}}^{(t)} = \left\| \frac{1}{m} \sum_{j=1}^m g^{(t)}(x_j) - \mathbb{E}[g^{(t)}] \right\|_2 \leq \sqrt{\frac{2 \log(1/\delta)}{m}}$$

for any iteration t .

LEMMA .2 (McDIARMID'S INEQUALITY). *Let $X_1, X_2, \dots, X_n$ be independent random variables taking values in sets $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_m$, respectively. Suppose a function $f : \prod_{i=1}^m \mathcal{X}_i \rightarrow \mathbb{R}$ satisfies the bounded difference property: there exist constants $c_1, c_2, \dots, c_n \geq 0$ such that for all $i \in [m]$,*

$$\sup_{x_1, \dots, x_m, x'_i} |f(x_1, \dots, x_i, \dots, x_m) - f(x_1, \dots, x'_i, \dots, x_m)| \leq c_i,$$

where the supremum is taken over all $x_j \in \mathcal{X}_j$ ($j \neq i$) and $x_i, x'_i \in \mathcal{X}_i$. Then for any $\lambda > 0$,

$$\Pr[f(X_1, \dots, X_m) - \mathbb{E}[f(X_1, \dots, X_m)] \geq \lambda] \leq \exp\left(-\frac{2\lambda^2}{\sum_{i=1}^n c_i^2}\right).$$

LEMMA .3 (JENSEN INEQUALITY). *For any convex function $\phi(\cdot)$ and integrable random variable X , we have $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$.*

In particular, for the ℓ_2 -norm (which is convex), it holds that $\|\mathbb{E}[X]\|_2 \leq \mathbb{E}[\|X\|_2]$; for a non-negative random variable X , we have $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$.

PROOF. of Lemma .1

Let $X_j := g^{(t)}(x_j) \in \mathbb{R}^N$ for each sample x_j in the private dataset at iteration t . By the algorithm design, $X_1, X_2, \dots, X_m$ are independent and identically distributed (i.i.d.) random vectors sampled from the true data population, and satisfy $\|X_j\|_2 \leq 1$ for all $j \in [m]$ according to the bound of gradient.

Denote the empirical mean as $\bar{X} = \frac{1}{m} \sum_{j=1}^m X_j$ and the true overall expectation as $\mu = \mathbb{E}[X_j]$. We aim to bound the ℓ_2 error between the empirical mean and true expectation, i.e., $\|\bar{X} - \mu\|_2$.

First, we define the function:

$$f(X_1, X_2, \dots, X_m) = \left\| \frac{1}{m} \sum_{j=1}^m X_j - \mu \right\|_2$$

For any $j \in [m]$, replacing X_j with X'_j (which also satisfies $\|X'_j\|_2 \leq 1$), by the reverse triangle inequality of the ℓ_2 norm, we have:

$$|f(\dots, X_j, \dots) - f(\dots, X'_j, \dots)| \leq \frac{1}{m} \|X_j - X'_j\|_2 \leq \frac{2}{m}.$$

Thus, f satisfies the bounded difference condition with constant $c_j = \frac{2}{m}$ for all $j \in [m]$.

By the one-sided McDiarmid's inequality (Lemma .2), for any $\lambda > 0$, we have:

$$\Pr(f - \mathbb{E}[f] \geq \lambda) \leq \exp\left(-\frac{2\lambda^2}{\sum_{j=1}^m c_j^2}\right) = \exp\left(-\frac{m\lambda^2}{2}\right).$$

Setting the right-hand side equal to δ solves to:

$$\lambda = \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

Therefore, with probability at least $1 - \delta$, it holds that:

$$f \leq \mathbb{E}[f] + \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

Next, we bound $\mathbb{E}[f]$ via Jensen's inequality (Lemma .3). We first expand $\mathbb{E}[f^2]$ using the independence of X_j :

$$\begin{aligned} \mathbb{E}[f^2] &= \mathbb{E}[\|\bar{X} - \mu\|_2^2] = \mathbb{E}\left[\left\| \frac{1}{m} \sum_{j=1}^m (X_j - \mu) \right\|_2^2\right] \\ &= \frac{1}{m^2} \sum_{j=1}^m \mathbb{E}[\|X_j - \mu\|_2^2], \end{aligned}$$

where cross terms vanish due to zero covariance between independent random variables. Since $\|X_j - \mu\|_2 \leq 2$, we have $\mathbb{E}[\|X_j - \mu\|_2^2] \leq 4$, which gives:

$$\mathbb{E}[f^2] \leq \frac{4}{m}.$$

By Jensen's inequality $\mathbb{E}[f]^2 \leq \mathbb{E}[f^2]$, we obtain:

$$\mathbb{E}[f] \leq \frac{2}{\sqrt{m}}.$$

Substituting (2) into (1), with probability at least $1 - \delta$, we have:

$$\|\bar{X} - \mu\|_2 \leq \frac{2}{\sqrt{m}} + \sqrt{\frac{2 \log(1/\delta)}{m}}.$$

For the standard setting in differential privacy where $\delta \ll 1$, we can rigorously absorb the constant term to get the concise bound:

$$\left\| \frac{1}{m} \sum_{j=1}^m g^{(t)}(x_j) - \mathbb{E}[g^{(t)}] \right\|_2 \leq \sqrt{\frac{2 \log(1/\delta)}{m}},$$

which completes the proof. $\square$

3.2 Error from DP Noise. We use a tree-based continual releasing mechanism in the algorithm to reduce noise aggregation by publishing interval statistical results. In error analysis, we use the worst-case error guarantee for it.

Under the tree-based continual release mechanism, the optimization update uses the noisy prefix increment

$$\Delta^{(t)} = \tilde{Z}^{(t)} - \tilde{Z}^{(t-1)},$$

where $\tilde{S}^{(t)}$ denotes the released noisy prefix sum.

We define the induced perturbation at iteration t as

$$e_{\text{est}}^{(t)} := \Delta^{(t)} - \bar{g}^{(t)}.$$

Since each prefix query is reconstructed from at most $\lceil \log T \rceil$ canonical tree nodes and every node contains independent Gaussian noise $\mathcal{N}(0, \sigma_{\text{tree}}^2 I)$, the perturbation satisfies

$$e_{\text{est}}^{(t)} \sim \mathcal{N}(0, \log T \cdot \sigma_{\text{tree}}^2 I_N).$$

3.3 Global error.

THEOREM .1 (CONVERGENCE ERROR BOUND OF DP-CRW). *Let $\mu = \mathbb{E}[g^{(t)}]$ be the true expected utility signal, and let $w^* \in \Delta_N$ denote the optimal probability distribution over the candidate set that maximizes the expected utility $w^* = \arg \max_{w \in \Delta_N} \langle w, \mu \rangle$, where $\Delta_N = \{w \in \mathbb{R}_{\geq 0}^N : \sum_{i=1}^N w_i = 1\}$ is the probability simplex. Let $w^{(t)}$ the weight vector at the t -th iteration, $g^{(t)}$ the true utility vector, N the size of the candidate set, m the number of samples in the private dataset, and T the total number of iteration rounds. Then, with probability at least $1 - \delta$, the average expected regret of the learned weight sequence satisfies*

$$\frac{1}{T} \sum_{t=1}^T \langle w^* - w^{(t)}, \mathbb{E}[g] \rangle \leq O\left(\sqrt{\frac{\log N}{T}} + \frac{1}{\sqrt{m}} + \frac{\sqrt{\log T}}{m\sqrt{\rho}}\right).$$

LEMMA .4 (HÖLDER'S INEQUALITY). *Let $p, q \in [1, \infty]$ be conjugate exponents satisfying $\frac{1}{p} + \frac{1}{q} = 1$. For any vectors $a, b \in \mathbb{R}^N$, we have*

$$|\langle a, b \rangle| \leq \|a\|_p \|b\|_q.$$

In particular, for the ℓ_1 - ℓ_∞ pair ($p = 1, q = \infty$), it holds that

$$|\langle a, b \rangle| \leq \|a\|_1 \|b\|_\infty.$$

PROOF. Based on the above derivations, we have completed the upper bound analysis for the three core error components of the noisy prefix sum $\tilde{Z}^{(t)} = \mathbb{E}[g^{(t)}] + e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)}$. On this basis, we substitute the above error terms into the continual weight refinement iteration process of the DP-CRW algorithm, to quantitatively analyze the end-to-end overall convergence error between the learned weight sequence after multiple rounds of exponential gradient descent iterations and the optimal weight distribution over the candidate set.

At iteration t , the algorithm updates the weights as

$$\log w_i^{(t)} \leftarrow \log w_i^{(t-1)} + \eta \tilde{Z}_i^{(t)}, \quad \forall i \in [N],$$

where $\tilde{Z}^{(t)} = (\tilde{Z}_1^{(t)}, \dots, \tilde{Z}_N^{(t)})$ is the noisy prefix sum vector released by the TreeAggregator mechanism. This shows that the proposed update rule can be interpreted as an Exponentiated Gradient (EG) procedure over the N -dimensional probability simplex $\Delta_N = \{w \in \mathbb{R}_{\geq 0}^N : \sum_{i=1}^N w_i = 1\}$.

To analyze the convergence of the update rule, we adopt the mirror descent framework with the negative entropy as the mirror map, i.e., $\psi(w) = \sum_{i=1}^N w_i \log w_i$, whose induced Bregman divergence is exactly the Kullback–Leibler (KL) divergence $D_{\text{KL}}(w \| v) = \sum_{i=1}^N w_i \log \frac{w_i}{v_i}$. This framework naturally aligns with the EG update, which operates on the probability simplex. At each iteration t , the algorithm uses the noisy prefix sum $\tilde{Z}^{(t)}$ as a surrogate gradient of the true expected utility vector $\mu = \mathbb{E}[g^{(t)}]$ to update the weight distribution $w^{(t)}$.

Under this formulation, we measure the algorithm's performance by the expected regret, which compares the cumulative expected utility achieved by the learned weight sequence $\{w^{(t)}\}_{t=1}^T$ against that of the best fixed optimal distribution $w^* \in \Delta_N$ in hindsight. Formally, the regret is defined as

$$\text{Regret}(T) = \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle,$$

where $w^* = \arg \max_{w \in \Delta_N} \langle w, \mu \rangle$ is the optimal distribution over the candidate set that maximizes the expected utility. The term $\langle w^* - w^{(t)}, \mu \rangle$ quantifies the suboptimality of the learned weight $w^{(t)}$ relative to w^* under the true expected utility $\mu = \mathbb{E}[g^{(t)}]$ at iteration t . Minimizing this regret ensures that the learned distribution $w^{(t)}$ progressively converges to the optimal distribution w^* , even with noisy gradient estimates.

We now derive the regret bound via the standard mirror descent regret guarantee for KL divergence [1]. For the EG update with learning rate η , the regret with respect to the noisy surrogate gradient $\tilde{Z}^{(t)}$ satisfies:

$$\sum_{t=1}^T \langle w^* - w^{(t)}, \tilde{Z}^{(t)} \rangle \leq \frac{D_{\text{KL}}(w^* \| w^{(0)})}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2.$$

For our algorithm, the initial weight $w^{(0)}$ is the uniform distribution over the candidate set, so $D_{\text{KL}}(w^* \| w^{(0)}) \leq \log N$ for any $w^* \in \Delta_N$. Substituting this into the bound gives:

$$\sum_{t=1}^T \langle w^* - w^{(t)}, \tilde{Z}^{(t)} \rangle \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2. \quad (1)$$

Next, we substitute the gradient decomposition $\tilde{Z}^{(t)} = \mu + e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)}$ into the left-hand side of 1:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle + \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)} \rangle \\ \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2. \end{aligned}$$

Rearranging terms, we obtain the bound for the target expected regret:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle &\leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 \\ &\quad - \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} + e_{\text{DP}}^{(t)} \rangle. \end{aligned} \quad (2)$$

We now bound the error terms on the right-hand side of (2) one by one. For any $w^*, w^{(t)} \in \Delta_N$, we have $\|w^* - w^{(t)}\|_1 \leq 2$ by the definition of the probability simplex. By Lemma .4 (ℓ_1 - ℓ_∞), for any error vector $e^{(t)}$, we have:

$$\left| \langle w^* - w^{(t)}, e^{(t)} \rangle \right| \leq \|w^* - w^{(t)}\|_1 \cdot \|e^{(t)}\|_\infty \leq 2\|e^{(t)}\|_\infty.$$

Alternatively, by the Cauchy-Schwarz inequality, we have:

$$\left| \langle w^* - w^{(t)}, e^{(t)} \rangle \right| \leq \|w^* - w^{(t)}\|_2 \cdot \|e^{(t)}\|_2 \leq \sqrt{2}\|e^{(t)}\|_2,$$

which aligns with the ℓ_2 bound of our pre-derived error terms. We use the ℓ_2 form below to match our previous lemmas.

Bound the empirical estimation error. By Lemma .1, for any iteration t , with probability at least $1 - \delta_1$, we have:

$$\|e_{\text{est}}^{(t)}\|_2 \leq \sqrt{\frac{2 \log(T/\delta_1)}{m}}.$$

We replace δ with δ_1/T to ensure the bound holds for all T iterations via the union bound. Summing over $t = 1$ to T , we get:

$$\left| \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} \rangle \right| \leq \sqrt{2} \sum_{t=1}^T \|e_{\text{est}}^{(t)}\|_2 \leq T \cdot \sqrt{\frac{4 \log(T/\delta_1)}{m}}. \quad (3)$$

Bound the DP noise perturbation.

To bound its effect on optimization, we first establish a dimension-free concentration inequality.

LEMMA .5 (GAUSSIAN CONCENTRATION FOR A LINEAR FUNCTIONAL OVER THE SIMPLEX). *Let $\xi \sim \mathcal{N}(0, \lambda^2 I_N)$, and let*

$$u = w^* - w^{(t)}, \quad w^*, w^{(t)} \in \Delta_N.$$

Then

$$\langle u, \xi \rangle \sim \mathcal{N}(0, \lambda^2 \|u\|_2^2).$$

Consequently, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$|\langle u, \xi \rangle| \leq \lambda \|u\|_2 \sqrt{2 \log \frac{2}{\delta}}.$$

Since $w^*, w^{(t)} \in \Delta_N$, we have $\|u\|_2 \leq 2$, and therefore

$$|\langle u, \xi \rangle| \leq 2\lambda \sqrt{2 \log \frac{2}{\delta}}.$$

The bound is independent of the ambient dimension N .

For our tree mechanism,

$$\lambda = \sqrt{\log T} \sigma_{\text{tree}}, \quad \sigma_{\text{tree}} = \frac{\sqrt{2 \log T}}{m\sqrt{\rho}}.$$

Applying Lemma .5 with failure probability δ_2/T yields that, for every $t \in [T]$,

$$\left| \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} \rangle \right| \leq 2\sqrt{\log T} \sigma_{\text{tree}} \sqrt{2 \log \frac{2T}{\delta_2}}$$

with probability at least $1 - \delta_2/T$.

By the union bound, the above inequality holds simultaneously for all $t \in [T]$ with probability at least $1 - \delta_2$.

Therefore,

$$\begin{aligned} \left| \sum_{t=1}^T \langle w^* - w^{(t)}, e_{\text{est}}^{(t)} \rangle \right| &\leq T \cdot 2\sqrt{\log T} \sigma_{\text{tree}} \sqrt{2 \log \frac{2T}{\delta_2}} \\ &= T \cdot 2\sqrt{\log T} \cdot \frac{\sqrt{2 \log T}}{m\sqrt{\rho}} \cdot \sqrt{2 \log \frac{2T}{\delta_2}} \\ &= \frac{4T \log T \sqrt{\log(2T/\delta_2)}}{m\sqrt{\rho}}. \end{aligned} \quad (4)$$

Bound the gradient norm term. Under the gradient's bound $\|g^{(t)}\|_2 \leq 1$, and with the DP noise bounded by a constant G_σ , we have $\|\tilde{Z}^{(t)}\|_\infty \leq G = 1 + G_\sigma$ for a universal constant G (since softmax outputs are in $[0, 1]$ and noise is bounded with high probability). Thus:

$$\eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 \leq \eta T G^2. \quad (5)$$

We now provide a coarse upper bound for G_σ to further characterize the magnitude of G .

We give the high-probability bound for the per-iteration noise norm, which aligns with the standard concentration inequality for Gaussian vectors and the union bound for T iterations:

LEMMA .6 (DP NOISE BOUND AT PER-ITERATION). *Let $\delta > 0$. For any iteration $t \in [T]$, with probability at least $1 - \delta/T$, the ℓ_2 norm of the tree-aggregation noise satisfies*

$$\|\xi^{(t)}\|_2 \leq \sqrt{\frac{2C^2 \log T}{m^2 \rho}} \cdot (\sqrt{2 \log(T/\delta)} + \sqrt{N}).$$

PROOF. Each coordinate of $\xi^{(t)}$ follows an independent Gaussian distribution $\mathcal{N}(0, \sigma_{\text{tree}}^2)$, so $\xi^{(t)}$ is an N -dimensional Gaussian vector with covariance matrix $\sigma_{\text{tree}}^2 I_N$. By the standard concentration inequality for the ℓ_2 norm of Gaussian vectors, for any $\gamma > 0$, we have

$$\Pr \left(\|\xi^{(t)}\|_2 \geq \sigma_{\text{tree}} \cdot (\sqrt{N} + \gamma) \right) \leq \exp \left(-\frac{\gamma^2}{2} \right).$$

For typical settings where $\log N < \log(T/\delta)$, we can absorb the dimension term into the logarithmic term, and set $\gamma = \sqrt{2 \log(T/\delta)}$ to make the right-hand side equal to δ/T . Substituting $\sigma_{\text{tree}} = \sqrt{\frac{2C^2 \log T}{m^2 \rho}}$ completes the proof. $\square$

Finally, we get:

$$\begin{aligned} \eta \sum_{t=1}^T \|\tilde{Z}^{(t)}\|_\infty^2 &\leq \eta T G^2 \\ &= \eta T \left(\sqrt{\frac{2C^2 \log T}{m^2 \rho}} (\sqrt{2 \log(T/\delta)} + \sqrt{N}) + 1 \right)^2. \end{aligned} \quad (6)$$

Combine all bounds and optimize the learning rate. Substitute (3), (4), (6) into (2). For the bounded utility setting where $\|g_{\text{emp}}^{(t)}\|_2 \leq 1$, and setting $\delta_1 = \delta_2 = \delta/2$ via the union bound to ensure the overall probability guarantee is at least $1 - \delta$, we obtain:

$$\begin{aligned} \sum_{t=1}^T \langle w^* - w^{(t)}, \mu \rangle &\leq \frac{\log N}{\eta} + \eta T G^2 \\ &\quad + T \sqrt{\frac{4 \log(2T/\delta)}{m}} \\ &\quad + \frac{4T \log T \sqrt{\log(2T/\delta)}}{m \sqrt{\rho}}. \end{aligned}$$

To minimize the bound, we optimize the learning rate η by balancing the first two terms. Taking the derivative with respect to η and setting it to zero gives the optimal learning rate:

$$\eta^* = \sqrt{\frac{\log N}{TG^2}}.$$

Substituting η^* into the bound, the first two terms become $2G\sqrt{T \log N}$.

Finally, we take the average over T iterations to get the average expected regret bound. Absorbing all constants into the big-O notation, we obtain the final convergence result:

$$\frac{1}{T} \sum_{t=1}^T \langle w^* - w^{(t)}, \mathbb{E}[g^{(t)}] \rangle \leq O\left(\sqrt{\frac{\log N}{T}} + \frac{1}{\sqrt{m}} + \frac{\log T}{m\sqrt{\rho}}\right),$$

which holds with probability at least $1 - \delta$. $\square$

4 Missing Experimental Details

4.1 Datasets. The ADULT dataset is obtained from the 1994 US Census database, consisting of 14 attributes (6 numerical attributes and 8 categorical attributes) and approximately 48,842 individual tuples. The core task of it is to classify the "income" of adults.

The HIGGS dataset is a high-energy physics dataset generated via Monte Carlo simulations of proton-proton collisions at the Large Hadron Collider. It comprises 28 attributes (21 numerical attributes and 7 categorical attributes) and around 81,000 samples, and serves as a common benchmark for high-dimensional data evaluation.

The HTRU dataset collects pulsar candidate observations from radio astronomy surveys, containing 9 attributes (8 numerical attributes and 1 categorical attributes) and approximately 17,898 individual signal samples. It is a typical benchmark for imbalanced classification and synthetic data validation.

The kdd-cup10 dataset originates from the "Educational Data Mining Challenge" held at the 2010 ACM SIGKDD International Conference. This dataset contains approximately 8 million interaction records between students and the tutoring system, involving 12 core attributes, including 6 numerical attributes and 6 categorical attributes. The core task of this subset is to predict the correctness of students' answers to questions.

The Shoppers dataset (full name: Online Shopper Behavior Dataset) is a benchmark dataset for shopping behavior focusing on the online retail scenario, containing 12,330 independent user session records. This dataset includes 10 numerical attributes and 8 categorical attributes, with the core target variable being "Revenue" (marking whether a session results in a purchase).

4.2 Prompt for APIs in generating candidate. Generate a synthetic tabular dataset of exactly 100,000 records in standard CSV format based on the full column names, data types, and value ranges. 1)

Restrict all numerical values to their designated ranges, and only use given categories for categorical columns. 2) Retain realistic semantic correlations between related features instead of generating columns separately. 3) Make every single record logically reasonable, without impossible attribute combinations. 4) Ensure broad coverage of each attribute's valid value space with diverse generated samples.

4.3 Metrics. We employ Wasserstein distance (W), Jensen-Shannon divergence (JSD) and ℓ_2 matrix three metrics to linearly evaluate the quality of our tabular synthetic data. Specifically, the Wasserstein distance quantifies the distributional similarity between synthetic and original data for continuous attributes, while the Jensen-Shannon divergence measures the distributional similarity for categorical attributes. The ℓ_2 matrix similarity metric provides a comprehensive evaluation by jointly considering all attribute types (both continuous and discrete) in the synthetic and original datasets. The functions are as follow:

$$\begin{aligned} W(P, Q) &= \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| dx \\ \text{JSD}(P||Q) &= \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q||M) \\ \ell_2 &= \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij} - b_{i,j}|^2} \end{aligned}$$

Where, P, Q are the probability distribution of original and synthetic datasets, $F_P(x), F_Q(x)$ are cumulative distribution functions of P, Q , $M = \frac{P+Q}{2}$ is the mixed distribution of P, Q , D_{KL} is the KL divergence, and m, n are the number of tuples and attributes.

4.4 Hyperparameters. Table 1 lists the main hyperparameters and the search spaces for each of them, which we recommend using.

Table 1: Model Performance Metrics

Hyperparameter	Search space
Learning rate	[0.0001, 0.0005]
Batch size	{256, 512, 1024}
Epsilon	{0.1, 0.5, 1.0, 10.0}
Delta	0.0001
Smoothing constan	$1e^{-6}$
Scheduler	cosine

REFERENCES

- [1] Shai Shalev-Shwartz. 2025. Online learning and online convex optimization. *Foundations and Trends® in Machine Learning* 4, 2 (2025), 107–194.